Midterm Walmart Case

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The Mess

- Of the 25,562 rows in our initial walmart_transactions.csv file some of our main issues were:
  - **886 Missing Timestamps:** Without a valid timestamp, a transaction can not be validly placed into the timeline. If we cannot place the day it occurred, we cannot assign it to either the baseline or surge period.
  - **30 Duplicate Rows:** Duplicate rows leads to double-counting the units that were actually sold and inflates sales volume.
  - **60 Distinct store_id for 12 stores:** Fragmentation across spacing, prefixes, and capitalization resulting in only 12 true stores.

The walmart_transaction csv file consisted of 25,562 rows. A few of the main issues with the data set was 886 missing timestamps meaning that we can't attribute a day to the transaction. 30 duplicate rows which would lead to double counting the units that were sold. And 60 distinct store_ids when there are only 12 true physical stores.

The Pipeline

- Dropped the 30 duplicate rows
- Cleaned unit_price category from walmart_transaction by stripping $, trailing whitespaces, and converted to float
- Standardized timestamps and dropped the 886 unparseable rows
- Cleaned and standardized store_id to follow FL-### format
- Used dictionary mapping to handle product and category variation
- Joined transitions with store metadata on store_id
- One decision we made: **Preserving Negative Quantities**
    - We made the decision to keep the negative quantities as they would be representative of refunds/ returns which is beneficial when calculating the true demand for items

For our cleaning pipeline we starting by removing duplicates and cleaning up the data types by removing dollar signs from prices, ensuring that quantities were stored as numbers and converting the timestamps into standard date format. As well as that, we also standardized product names, categories, and store IDs so that when merging the two tables we would be able to match them according to their stores

An important decision we made was the decision to keep the negative quantities. We felt that keeping them would be representative of the amount of return or refunds which would be important in our analysis as returns impact how much inventory a store has, as well as get a more accurate picture of demand.

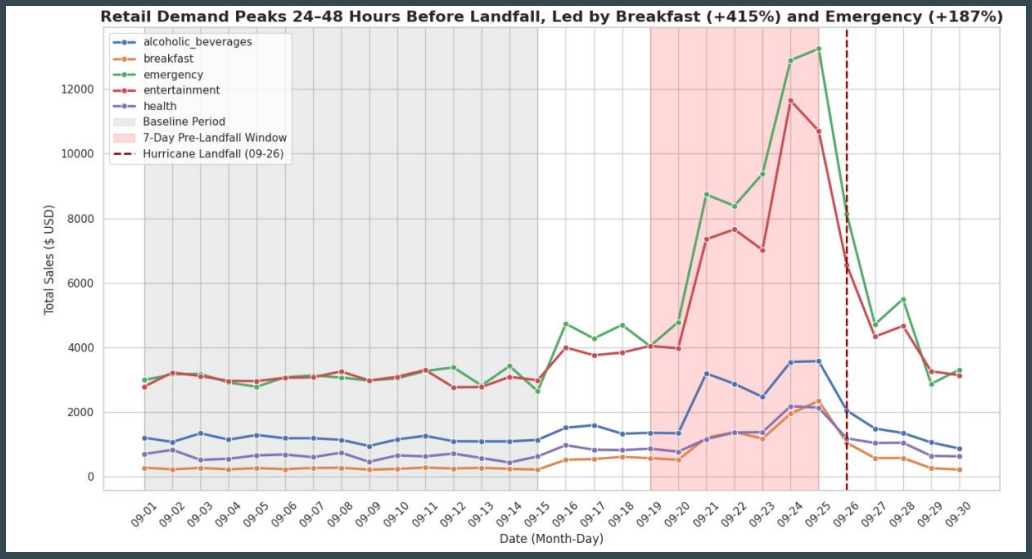
Findings Q1

Question 1: Demand Surge Identification

- The top 5 product categories that we found were: Breakfast, Emergency, Entertainment, Alcoholic Beverages, and Health. While Breakfast had the highest relative surge, of +415.4%, Emergency had the largest increase in daily revenue, with $5720.42.
- We picked our baseline period to be September 1-15 since we'd be able to get a better picture of normal sales throughout the week, and our 7-day surge window was September 19-25.
- We found that sales accelerated starting the 21st and peaked between the 24th and 25th before plunging on September 26th (landfall).

For Question 1, we compared the average daily sales revenue during our baseline and surge period to determine which product categories experience the largest increases. We saw that the top 5 categories were breakfast, emergency, entertainment, alcoholic beverages, and health. Breakfast had the highest relative surge of +415.4% but emergency had the largest increase in daily revenue of about 5,700. Our baseline was sept 1 - 15 and our 7 day surge window was the 19th -25 and we saw that sales peaked between the 24th and 25th before plunging on the 26th.

Q1 Chart



Looking at the chart we can see that starting around the 21st is when we first see the spike in sales and they start reach a peak around the 24th and 25th of september. After that when the landfall hits on the 26th we see a generally drastic steep decline in sales.

While on the chart we do see emergency and entertainment dominating the chart in total sales, going over $10,000 in sales, it was actually breakfast that had the highest surge multiple of 5.15x. On the chart it looks small because it’s normal baseline was pretty modest starting around $250 per day.

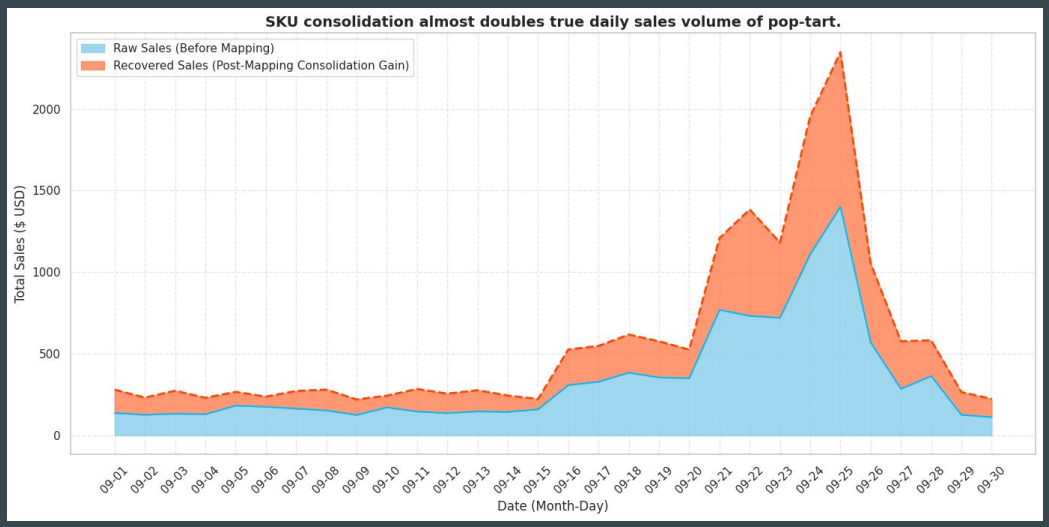
Findings Q2

Question 2: The Pop-Tarts Effect (SKU Consolidation)
After standardizing all Pop-Tart SKU variants into a single product, what is the true daily sales volume? How does the uncleaned (fragmented) view compare to the cleaned (consolidated) view? What business decisions would differ between the two views?

- Before pop-tart consolidation, there were 4 fragmented SKU values all representing the same product. The consolidated view shows a total of 5,027 units sold in September 2024, generating $17,587.35 in true revenue. The uncleaned view captured only 2,903 units.
- Although the general trends were similar, if not the same, for both before and after mapping, consolidation almost doubled the amount of daily sales, which would be beneficial for decision makers to know. It would highly affect reports of daily sales, which is important to know to determine how much of each product to stock.

- For Q2, we built mapping rules that consolidated 45 unique SKUs to 20. Each SKU was mapped based on shared key words that allowed us to combine by product_name. After consolidating product names, we were able to evaluate how before and after mapping affected performance reports.
- We noticed that by failing to clean and group the variations, **2,124 units (approx. 42.2% of the actual total volume) were completely hidden** across fragmented SKUs.
- It would highly affect reports of daily sales, which is important to know to determine how much of each product to stock, prevent bottlenecking of products coming from distribution centers, and allow decision makers to properly allocate shelf space

Q2 Chart


SKU consolidation almost doubles true daily sales volume of pop-tart.
Raw Sales (Before Mapping)
Recovered Sales (Post-Mapping Consolidation Gain)
Total Sales ($ USD)
0
500
1000
1500
2000
09-01
09-02
09-03
09-04
09-05
09-06
09-07
09-08
09-09
09-10
09-11
09-12
09-13
09-14
09-15
09-16
09-17
09-18
09-19
09-20
09-21
09-22
09-23
09-24
09-25
09-26
09-27
09-28
09-29
09-30
Date (Month-Day)

Findings Q3

Question 3: Weather Correlation Analysis

Using weather data pulled from your chosen free API, how do specific weather indicators (wind speed, precipitation, barometric pressure) correlate with daily sales volume across categories? Is there a detectable lag between weather severity escalation and purchasing spikes?

- Sales increased approximately two days before stronger winds and lower barometric pressure, suggesting customers prepared for the approaching storm.
- The strongest relationship was with wind speed (r = 0.803), indicating higher sales tended to precede stronger winds.
- Lower atmospheric pressure also showed a strong relationship (r = -0.728), suggesting purchasing increased before storm conditions intensified.
- Precipitation showed weaker relationships, suggesting rainfall was less closely associated with advance purchasing.
- Sales peaked before Hurricane Helene's landfall and declined as severe weather arrived.
- Overall, the 2-day delay between purchasing spikes and worsening weather suggests customers anticipated the storm and stocked up in advance, although these correlations do not prove causation.

For Question 3, we analyzed how wind speed, precipitation, and barometric pressure correlated with daily sales across Walmart stores during Hurricane Helene.

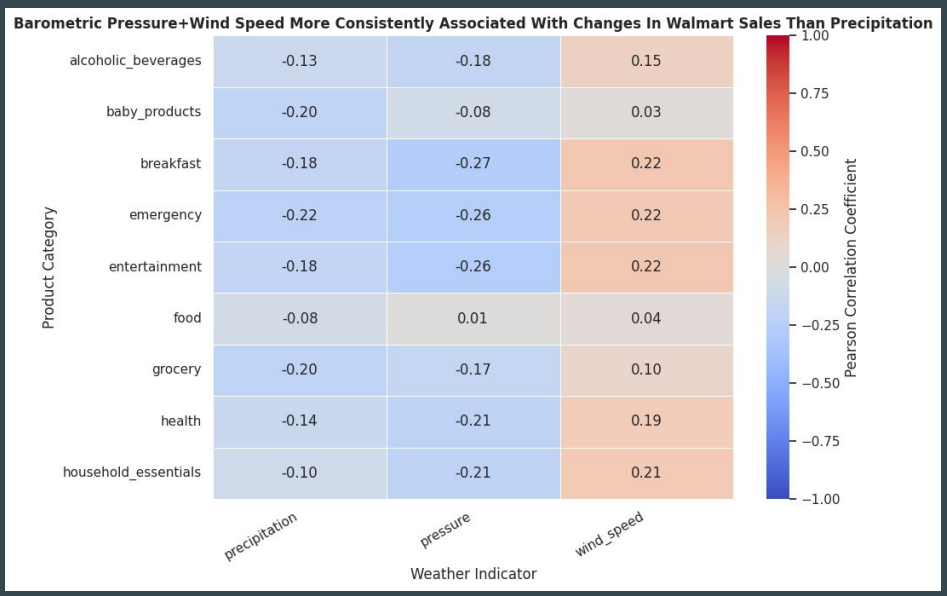
Our analysis found that sales increased approximately two days before severe weather conditions intensified, suggesting customers were preparing for the approaching hurricane.

Wind speed showed the strongest correlation, at 0.803, meaning higher sales tended to occur before stronger winds. Barometric pressure also showed a strong negative correlation of −0.728, meaning sales increased ahead of the lower pressure associated with the storm.

Precipitation had a weaker relationship with sales, suggesting that customers were responding more to the approaching hurricane than to rainfall itself.

Overall, our findings suggest that customers anticipated the storm and stocked up about two days in advance. However, while these correlations show a strong relationship between weather conditions and purchasing behavior, they don't necessarily prove causation.

Q3 Chart



This heatmap shows how weather conditions correlated with Walmart sales across different product categories.

Wind speed had positive correlations across all categories, especially breakfast, emergency supplies, and entertainment.

Barometric pressure and precipitation showed mostly negative correlations, meaning sales tended to increase as pressure and rainfall decreased.

Overall, the correlations were relatively weak, but wind speed and pressure showed consistent patterns across categories.

Findings Q4

Question 4: Store-Level Geographic Patterns

Do all Florida store locations experience the same surge patterns, or do stores closer to the predicted hurricane path show earlier and larger spikes? Identify any stores that appear to be outliers.

- **Geographic patterns:** All 12 Florida stores experienced sales increases of 125.05%–173.00% before Hurricane Helene, although surge magnitudes varied.
- **Proximity to hurricane path:** Gainesville (FL-078), closer to Helene's landfall, experienced a 135.53% surge, while Miami (FL-330), farther away, recorded the largest increase at 173.00%.
- **Surge timing:** Kissimmee (FL-287) experienced the earliest spike on September 19, seven days before landfall. Miami and Tallahassee spiked on September 20, while the other nine stores first spiked on September 21.
- **Peak sales:** Most stores reached peak sales on September 24–25, with several reaching approximately 300–420% of baseline daily sales.
- **Outliers:** Miami (FL-330) exceeded the upper IQR threshold with the largest average surge (173.00%), while Orlando (FL-239) fell below the lower threshold with the smallest (125.05%).
- **Conclusion:** Stores closer to the hurricane's landfall did not consistently experience earlier or larger surges (r = 0.338 for distance versus spike timing). Overall, hurricane preparation increased sales across Florida regardless of proximity.

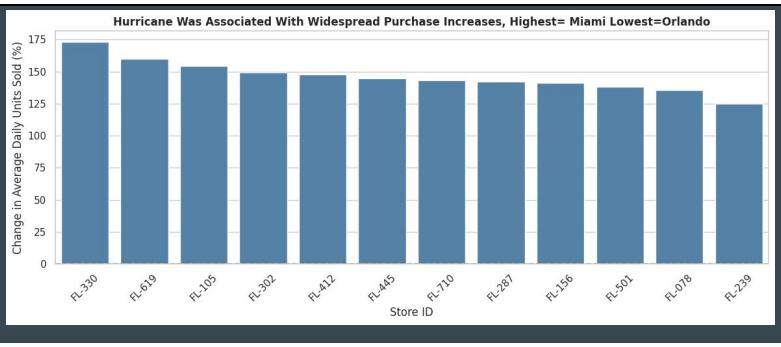
All 12 Florida stores experienced sales increases before Hurricane Helene, ranging from 125% to 173%.

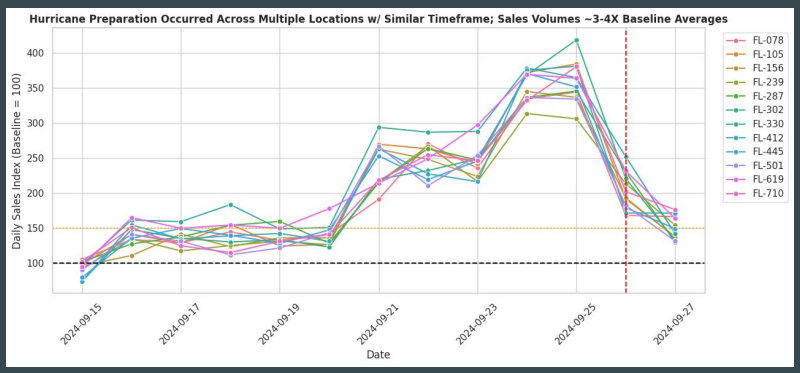
Miami had the largest average surge at 173%, despite being farther from the hurricane's landfall, while Orlando had the smallest at 125%.

Kissimmee experienced the earliest spike on September 19, followed by Miami and Tallahassee on September 20. Most other stores began surging on September 21 and peaked around September 24–25.

Overall, proximity to the hurricane did not necessarily mean larger or earlier sales increases. The weak correlation of 0.338 between distance and spike timing suggests that other factors may have influenced when customers began preparing.

Q4 Chart





The top graph compares sales increases across all 12 Florida stores. Miami had the largest average surge at 173%, while Orlando had the smallest at 125%.

The bottom graph shows when customers began stocking up. Kissimmee experienced the earliest spike on September 19, followed by Miami and Tallahassee on September 20, while most other stores followed on September 21.

Sales peaked around September 24–25, just before Hurricane Helene made landfall on September 26. Overall, stores farther from landfall still experienced major increases in sales, showing that proximity did not consistently determine purchasing behavior.

Findings Q5

Question 5: Signal vs. Artifact — The Board Games Investigation

The board games/entertainment category shows a +250% sales increase. Using the SQLite inventory and sales database, investigate whether this is a genuine demand signal or a data artifact (e.g., SKU miscoding, category lumping, inventory reporting lag). Present your evidence and make a recommendation: should Walmart stock extra board games for the next storm?

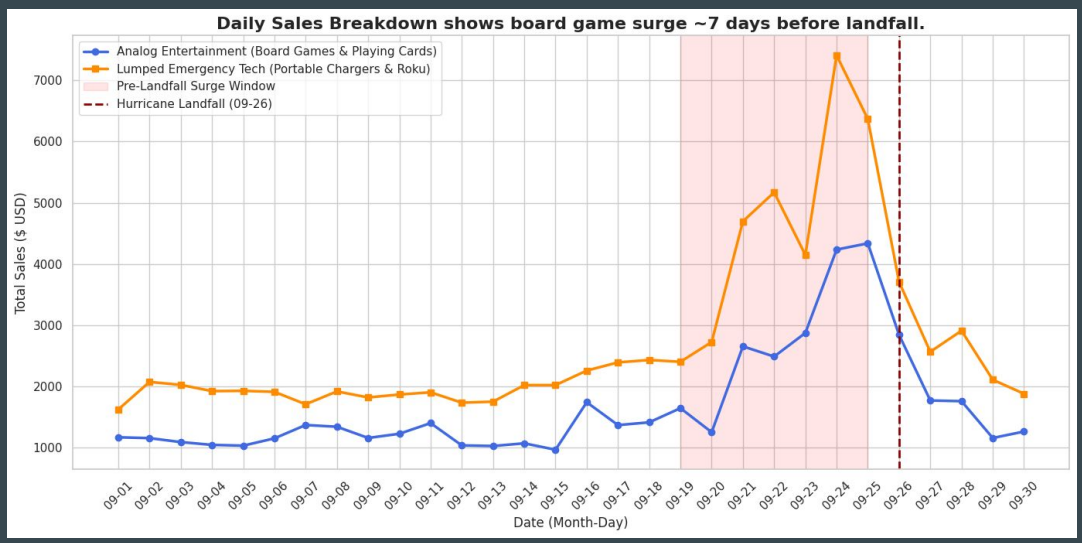
- **Genuine Demand:** Data shows that the surge is a genuine demand signal, as after removing category lumping, board games and cards saw a 294% increase in daily sales on September 25.
- Based on the surge window and the sales spike, Walmart should stock extra board games for the next storm about 7-10 days before the landfall to capture the highest revenue margin.

Yes, Walmart should definitely stock extra board games and playing cards for the next storm.

1. Genuine Analog Demand Surge: Even when removing lumped electronics, actual Board Games and Cards (Monopoly and playing cards) alone experienced a massive ~294% surge in daily sales, leaping from a baseline of 1100 dollars per day to 4300 dollars per day on September 25. This shows a powerful demand signal from families preparing for multi-day power outages with non-electric, analog entertainment.
2. High Stockout Risk: Analog board games are normally slow-moving inventory. A sudden ~300% spike in demand can quickly empty store shelves. Increasing safety stock of family-oriented board games and low-cost card decks 7–10 days before forecasted landfalls will capture this high-margin revenue and keep customers satisfied.
3. Cross-Merchandising Opportunity: The visual evidence shows that emergency electronics (like portable chargers) and board games spike in tandem during the exact same pre-landfall window. Walmart should utilize this behavior by building endcap displays that co-locate portable power accessories alongside analog board games, flashlights, and

1. emergency candles under a "Power Outage Preparedness" theme.

Q5 Chart


Daily Sales Breakdown shows board game surge ~7 days before landfall.
Analog Entertainment (Board Games & Playing Cards)
Lumped Emergency Tech (Portable Chargers & Roku)
Pre-Landfall Surge Window
Hurricane Landfall (09-26)
Total Sales ($ USD)
Date (Month-Day)

Recommendations

- **Increase emergency inventory** 7 days before predicted landfall to prepare for purchasing spikes observed as early as September 19.
- **Prioritize high-demand stores:** Miami (173.00%), Tallahassee (160.05%), and Tampa (154.53%) experienced the largest average sales surges.
- **Schedule additional deliveries** 5–7 days before landfall to maintain inventory availability during peak purchasing periods.
- **Increase staffing** beginning 5 days before landfall, with additional support during the final 48 hours to accommodate higher transaction volumes and faster shelf restocking.

Based on our findings, we recommend increasing emergency inventory at least seven days before predicted landfall, since purchasing spikes began earlier than expected.

Stores like Miami, Tallahassee, and Tampa should receive priority because they experienced the largest sales increases.

We also recommend scheduling deliveries five to seven days before landfall and increasing staffing as purchasing activity intensifies.

What Broke

- While our pipeline didn't explicitly have any red fatal errors, there were silent data errors that were overlooked in our initial process.
  - Initial usage of .str.contains() for sku mapping overlooked reverse string names such as (Bread - White and TP 12 Pack) leaving it unmerged
  - First pass of mapping left electronics split from entertainment and snacks split from food.
- Overall we didn't experience any major breaks or merge conflicts when working on this project.

Overall in our project we didn't experience any major crashes or breaks however there were slight silent data eros that could've distorted our numbers and final results. Our substrings rules initially overlooked reversed product names such as 'Bread - White' and 'TP 12 Pack' leaving them unconsolidated until we added an additional SKU remap. Secondly, electronics and snacks were initially split from entertainment and food which would've distorted entertainment's true surge rank.